

COS 433: Cryptography

Announcements:

- Problem Set 2 due today
- Problem Set 3 released today, due October 28th

One-Way Functions

Theorem [Håstad-Impagliazzo-Levin-Luby '90] (we will **partially** prove):

If **OWFs exist**, then **PRGs exist!**

PRGs from One-Way Permutations

Theorem [Yao '82, Goldreich-Levin '89]:

If **One-way permutations exist**, then **PRGs exist!**

Goldreich-Levin Theorem

Theorem [Goldreich-Levin '89]:

For every OWF f , for all poly-time A ,

$$\Pr_{x,r \leftarrow \{0,1\}^\lambda} [A(f(x), r) = \langle x, r \rangle] = \frac{1}{2} + \text{negl}(\lambda)$$

Goldreich-Levin Theorem

Proof of GL theorem:



Proof is tricky

Goldreich-Levin Theorem

Proof of GL theorem:

Proof by contradiction. Suppose that polynomial-time A satisfies

$$\Pr_{x,r \leftarrow \{0,1\}^\lambda} [A(f(x), r) = \langle x, r \rangle] = \frac{1}{2} + \epsilon(\lambda)$$

Eventual goal: construct poly-time A' such that

$$\Pr_{x \leftarrow \{0,1\}^\lambda} [A'(f(x)) = x] = \epsilon'(\lambda)$$

Goldreich-Levin Theorem

Proof of GL theorem:

Proof by contradiction.

First, an easy case. Suppose that A is **perfect**:

$$\Pr_{x,r \leftarrow \{0,1\}^\lambda} [A(f(x), r) = \langle x, r \rangle] = 1$$

Claim: easy to invert f with probability 1.

How? Given $y = f(x)$, call $A(y, e_1) \rightarrow x_1, \dots, A(y, e_\lambda) \rightarrow x_\lambda$. **All correct with prob 1.**

Goldreich-Levin Theorem

Proof of GL theorem:

Proof by contradiction.

Next, a **medium** case. Suppose that A is **very good**:

$$\Pr_{x, r \leftarrow \{0,1\}^\lambda} [A(f(x), r) = \langle x, r \rangle] = \frac{3}{4} + \epsilon(\lambda)$$

Problem: A may be **completely useless** for certain x

Solution: Averaging argument. For every x , let $p_x = \Pr_{r \leftarrow \{0,1\}^\lambda} [A(f(x), r) = \langle x, r \rangle]$

Goldreich-Levin Theorem

Proof of GL theorem:

Proof by contradiction.

Next, a **medium** case. Suppose that A is **very good**:

$$\mathbf{E}_{x \leftarrow \{0,1\}^\lambda} [p_x] = \frac{3}{4} + \epsilon(\lambda)$$

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Goldreich-Levin Theorem

Proof of GL theorem:

Proof by contradiction.

Next, a **medium** case. Suppose that A is **very good**:

$$\mathbf{E}_{x \leftarrow \{0,1\}^\lambda} [p_x] = \frac{3}{4} + \epsilon(\lambda)$$

Claim: For at least an $\epsilon/2$ fraction of x , $p_x \geq \frac{3}{4} + \epsilon/2$ (call these “good x ”)

Note: adversary doesn't know which x have this property.

Goldreich-Levin Theorem

Proof of GL theorem:

Proof by contradiction.

Next, a **medium** case. Suppose that A is **very good**:

$$\mathbf{E}_{x \leftarrow \{0,1\}^\lambda} [p_x] = \frac{3}{4} + \epsilon(\lambda)$$

Claim: For at least an $\epsilon/2$ fraction of x , $p_x \geq \frac{3}{4} + \epsilon/2$ (call these “good x ”)

We will define an A' that inverts every **good x** whp.

Goldreich-Levin Theorem

Proof of GL theorem:

Goal: define $A'(y)$ such that for all **good** x ,

$$\Pr[A'(f(x)) = x] = 1 - o(1)$$

Old way: given $y = f(x)$, call $A(y, e_1) \rightarrow x_1, \dots, A(y, e_\lambda) \rightarrow x_\lambda$.

No good! **Every single one of these answers might be wrong!**

Goldreich-Levin Theorem

Proof of GL theorem:

Goal: define $A'(y)$ such that for all **good** x ,

$$\Pr[A'(f(x)) = x] = 1 - o(1)$$

Ideas?

Goldreich-Levin Theorem

Proof of GL theorem:

Goal: define $A'(y)$ such that for all **good** x ,

$$\Pr[A'(f(x)) = x] = 1 - o(1)$$

Let's assume A is **deterministic** (wlog, also doesn't really matter).

- then for every (x, r) , $A(y = f(x), r)$ is either right or wrong.
- for every **good** x , A_y is right on at least $\frac{3}{4} + \frac{\epsilon}{2}$ fraction of r .

Goldreich-Levin Theorem

Proof of GL theorem:

Goal: define $A'(y)$ such that for all **good** x ,

$$\Pr[A'(f(x)) = x] = 1 - o(1)$$

Big idea: **error correction!**

Goldreich-Levin Theorem

Proof of GL theorem:

Algorithm $A'(y)$:

- Sample strings $r_1, r_2, \dots, r_n$ for $n = O\left(\frac{\lambda}{\epsilon^2}\right)$
- For each i , we will come up with a guess for x_i
 1. For all j , compute $A_y(r_j)$ and $A_y(r_j \oplus e_i)$
 2. Compute $x_i^{(j)} = A_y(r_j) \oplus A_y(r_j \oplus e_i)$ // this = x_i if A_y is right twice.
 3. Guess $x_i = \text{Majority}(x_i^{(1)}, x_i^{(2)}, \dots, x_i^{(n)})$
- Output guesses.

Goldreich-Levin Theorem

Key point: fix an x . For all i, j :

$$\Pr_{r_j} \left[x_i^{(j)} = x_i \right] \geq 2 \cdot p_x - 1$$

$$\geq 1/2 + \epsilon$$

Algorithm $A'(y)$:

- Sample strings $r_1, r_2, \dots, r_n$ for $n = O\left(\frac{\lambda}{\epsilon^2}\right)$
- For each i , we will come up with a guess for x_i
 1. For all j , compute $A_y(r_j)$ and $A_y(r_j \oplus e_i)$
 2. Compute $x_i^{(j)} = A_y(r_j) \oplus A_y(r_j \oplus e_i)$ // this = x_i if A_y is doubly right.
 3. Guess $x_i = \text{Majority}(x_i^{(1)}, x_i^{(2)}, \dots, x_i^{(n)})$
- Output guesses.

Moreover, $x_i^{(1)}, \dots, x_i^{(n)}$ are independent! Therefore (see pset 0 Problem 3c),

$$\Pr_{r_1, \dots, r_n} \left[\frac{1}{n} \cdot \# \left\{ j : x_i^{(j)} = x_i \right\} \leq 1/2 \right] \leq \frac{1}{n \cdot \epsilon^2}$$

$$\leq \frac{1}{100 \cdot \lambda}$$

Goldreich-Levin Theorem

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$$\Pr_{r_j} \left[x_i^{(j)} = x_i \right] \geq 2 \cdot p_x - 1 \\ \geq 1/2 + \epsilon$$

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- Output guesses.

Moreover, $x_i^{(1)}, \dots, x_i^{(n)}$ are independent! Therefore (see pset 0 Problem 3c),

$$\Pr_{r_1, \dots, r_n} \left[i^{\text{th}} \text{ guess is wrong} \right] \leq \frac{1}{n \cdot \epsilon^2} \\ \leq \frac{1}{100 \cdot \lambda}$$

Goldreich-Levin Theorem

Key point: fix an x . For all i, j :

$$\Pr_{r_j} [x_i^{(j)} = x_i] \geq 2 \cdot p_x - 1 \\ \geq 1/2 + \epsilon$$

Conclusion:

$$\Pr_{r_1, \dots, r_n} [\text{any guess is wrong}] \leq \frac{1}{100}$$

Algorithm $A'(y)$:

- Sample strings $r_1, r_2, \dots, r_n$ for $n = O\left(\frac{\lambda}{\epsilon^2}\right)$
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 3. Guess $x_i = \text{Majority}(x_i^{(1)}, x_i^{(2)}, \dots, x_i^{(n)})$
- Output guesses.

Goldreich-Levin Theorem

Proof of GL theorem:

Proof by contradiction.

Finally, the **general** case. Suppose that A is **barely non-trivial**:

$$\Pr_{x,r \leftarrow \{0,1\}^\lambda} [A(f(x), r) = \langle x, r \rangle] = \frac{1}{2} + \epsilon(\lambda)$$

Averaging: For at least an $\epsilon/2$ fraction of x , $p_x \geq \frac{1}{2} + \epsilon/2$ (call these “decent x ”)

Goldreich-Levin Theorem

Proof of GL theorem:

Goal. Find A' such that for all **decent** x ,

$$\Pr[A'(f(x)) = x] = 1 - o(1)$$

This is impossible!

Goldreich-Levin Theorem

Proof of GL theorem:

Goal: define $A'(y)$ such that for all **decent** x ,

$$\Pr[A'(f(x)) = x] = 1 - o(1)$$

Example: suppose that

- $y = f(x_0) = f(x_1)$
- $A(y, r) = \langle x_0, r \rangle$ if $\langle x_0, r \rangle = \langle x_1, r \rangle$
- $A(y, r)$ even split between $\langle x_0, r \rangle$ and $\langle x_1, r \rangle$ otherwise.

Claim: $p_{x_1} = p_{x_2} = \frac{3}{4}$, but
 $A'(y)$ cannot tell x_1 from x_2

Goldreich-Levin Theorem

Proof of GL theorem:

Goal. Given A' , show that for all **decent** x ,

$$\Pr[A'(f(x)) = x] = 1 - o(1)$$

Can't decode a **unique** x from the available information $\{A(y, r)\}_r$

Goldreich-Levin Theorem

Proof of GL theorem:

Goal: define $A'(y)$ such that for all **decent** x ,

$$\Pr[A'(f(x)) = x] = \frac{1}{\text{poly}(\lambda)}$$

Magic: we can narrow down to a **few possibilities** for x

Goldreich-Levin Theorem

Proof of GL theorem:

Algorithm $A'(y)$:

- Sample strings $s_1, s_2, \dots, s_t$ and bits $a_1, a_2, \dots, a_t$ for $n = O\left(\frac{\lambda}{\epsilon^2}\right)$ and $t = \log(n + 1)$
- For $1 \leq j \leq n$, define $r_j = \sum_{\ell=1}^t j_\ell \cdot s_\ell$ and $b_j = \sum_{\ell=1}^t j_\ell \cdot a_\ell$ (bit decomp of j)
- For each i , we will come up with a guess for x_i
 1. For all j , compute $A_y(r_j \oplus e_i)$
 2. Compute $x_i^{(j)} = b_j \oplus A_y(r_j \oplus e_i)$ // this = x_i if A_y and b_j are right.
 3. Guess $x_i = \text{Majority}(x_i^{(1)}, x_i^{(2)}, \dots, x_i^{(n)})$
- Output guesses.

Goldreich-Levin Theorem

Key point: fix an x , $s_1, \dots, s_t$. Then,

$$\Pr_{a_1, \dots, a_t} [a_\ell = \langle x, s_\ell \rangle \text{ for all } \ell] = 2^{-t} = \frac{1}{n}$$

Algorithm $A'(y)$:

- Sample strings $s_1, s_2, \dots, s_t$ and bits $a_1, a_2, \dots, a_t$ for $n = O\left(\frac{\lambda}{\epsilon^2}\right)$ and $t = \log n$
- For $1 \leq j \leq n$, define $r_j = \sum_{\ell=1}^t j_\ell \cdot s_\ell$ and $b_j = \sum_{\ell=1}^t j_\ell \cdot a_\ell$ (bit decomp of j)
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 3. Guess $x_i = \text{Majority}(x_i^{(1)}, x_i^{(2)}, \dots, x_i^{(n)})$
- Output guesses.

When this event occurs, we then have for all j :

$$b_j = \sum_{\ell=1}^t j_\ell \cdot a_\ell = \sum_{\ell=1}^t j_\ell \cdot \langle x, s_\ell \rangle = \langle x, r_j \rangle$$

Goldreich-Levin Theorem

Analysis: fix x , i and condition on event $E = \{a_\ell = \langle x, s_\ell \rangle \text{ for all } \ell\}$. Then, for all j :

$$\Pr \left[x_i^{(j)} = x_i \mid E \right] = p_x \geq 1/2 + \epsilon/2$$

Algorithm $A'(y)$:

- Sample strings $s_1, s_2, \dots, s_t$ and bits $a_1, a_2, \dots, a_t$ for $n = O\left(\frac{\lambda}{\epsilon^2}\right)$ and $t = \log n$
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 3. Guess $x_i = \text{Majority}(x_i^{(1)}, x_i^{(2)}, \dots, x_i^{(n)})$
- Output guesses.

Moreover, $r_1, \dots, r_n$ are pairwise independent! That is, $\forall j, j'$: $(r_j, r_{j'})$ are independent.

$$r_1 = s_1 \quad r_2 = s_2 \quad r_3 = s_1 \oplus s_2 \quad r_4 = s_3 \quad r_7 = s_3 \oplus s_2 \oplus s_1$$

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 3. Guess $x_i = \text{Majority}(x_i^{(1)}, x_i^{(2)}, \dots, x_i^{(n)})$
- Output guesses.

Moreover, $r_1, \dots, r_n$ are pairwise independent! That is, $\forall j, j'$: $(r_j, r_{j'})$ are independent.

Therefore, (conditioned on E) the $x_i^{(j)} = \langle x, r_j \rangle \oplus A_y(r_j \oplus e_i)$ are pairwise independent

Goldreich-Levin Theorem

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Algorithm $A'(y)$:

- Sample strings $s_1, s_2, \dots, s_t$ and bits $a_1, a_2, \dots, a_t$ for $n = O\left(\frac{\lambda}{\epsilon^2}\right)$ and $t = \log n$
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- Output guesses.

Moreover, $x_i^{(1)}, \dots, x_i^{(n)}$ are pairwise independent! Therefore (see pset 0 Problem 3c),

$$\Pr_{\substack{s_1, \dots, s_t \\ a_1, \dots, a_t}} \left[\frac{1}{n} \cdot \# \left\{ j : x_i^{(j)} = x_i \right\} \leq \frac{1}{2} \mid E \right] \leq \frac{1}{n \cdot \epsilon^2} \leq \frac{1}{100 \cdot \lambda}$$

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Analysis: fix x , i and condition on event $E = \{a_\ell = \langle x, s_\ell \rangle \text{ for all } \ell\}$. Then, for all j :

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 3. Guess $x_i = \text{Majority}(x_i^{(1)}, x_i^{(2)}, \dots, x_i^{(n)})$
- Output guesses.

Moreover, $x_i^{(1)}, \dots, x_i^{(n)}$ are pairwise independent! Therefore (see pset 0 Problem 3c),

$$\Pr_{\substack{s_1, \dots, s_t \\ a_1, \dots, a_t}} \left[i^{\text{th}} \text{ guess is wrong} \mid E \right] \leq \frac{1}{n \cdot \epsilon^2} \leq \frac{1}{100 \cdot \lambda}$$

Goldreich-Levin Theorem

Analysis: fix x , i and condition on event $E = \{a_\ell = \langle x, s_\ell \rangle \text{ for all } \ell\}$. Then, for all j :

$$\Pr \left[x_i^{(j)} = x_i \mid E \right] = p_x \geq 1/2 + \epsilon/2$$

Algorithm $A'(y)$:

- Sample strings $s_1, s_2, \dots, s_t$ and bits $a_1, a_2, \dots, a_t$ for $n = O\left(\frac{\lambda}{\epsilon^2}\right)$ and $t = \log n$
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- Output guesses.

Moreover, $x_i^{(1)}, \dots, x_i^{(n)}$ are pairwise independent! Therefore (see pset 0 Problem 3c),

$$\Pr_{\substack{s_1, \dots, s_t \\ a_1, \dots, a_t}} [\text{any guess is wrong} \mid E] \leq \frac{1}{100}$$

Goldreich-Levin Theorem

Analysis: fix x , i and condition on event $E = \{a_\ell = \langle x, s_\ell \rangle \text{ for all } \ell\}$. Then, for all j :

$$\Pr \left[x_i^{(j)} = x_i \mid E \right] = p_x \geq 1/2 + \epsilon/2$$

Algorithm $A'(y)$:

- Sample strings $s_1, s_2, \dots, s_t$ and bits $a_1, a_2, \dots, a_t$ for $n = O\left(\frac{\lambda}{\epsilon^2}\right)$ and $t = \log n$
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 3. Guess $x_i = \text{Majority}(x_i^{(1)}, x_i^{(2)}, \dots, x_i^{(n)})$
- Output guesses.

Moreover, $x_i^{(1)}, \dots, x_i^{(n)}$ are pairwise independent! Therefore (see pset 0 Problem 3c),

$$\Pr_{\substack{s_1, \dots, s_t \\ a_1, \dots, a_t}} [\text{all guesses are right}] \geq \frac{1}{n} \cdot \frac{99}{100}$$

Goldreich-Levin Theorem

Proof of GL theorem:

Conclusion: for all decent x ,

$$\Pr[A'(f(x)) = x] \geq \frac{99}{100 \cdot n}$$

and so

$$\Pr_{\mathbf{x}}[A'(f(\mathbf{x})) = \mathbf{x}] \geq \frac{99 \cdot \epsilon}{200 \cdot n} = \text{poly}(\epsilon, 1/\lambda)$$

Goldreich-Levin Theorem

Theorem [Goldreich-Levin '89]:

For every OWF f , for all poly-time A ,

$$\Pr_{x,r \leftarrow \{0,1\}^\lambda} [A(f(x), r) = \langle x, r \rangle] = \frac{1}{2} + \text{negl}(\lambda)$$

Next time: number theoretic cryptography